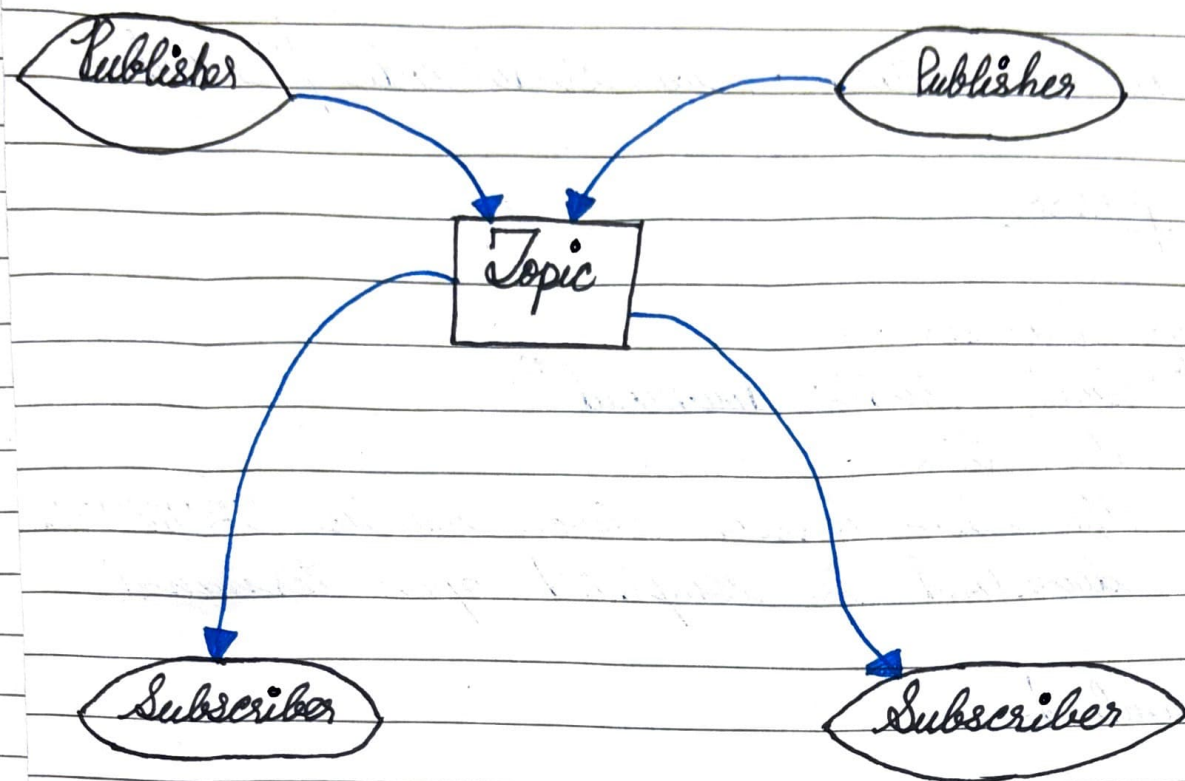




Nodes: These are the individual pieces of code or processes within a robotic system that perform specific tasks, such as navigation or reading sensor data.

Publishers: These are nodes responsible for sending information like sensor readings or status updates, across ROS2 network to the rest of the robot.

Subscribers: These are the nodes responsible for receiving data published by other ~~robot~~ nodes. For example, a navigation node might subscribe to data from a laser scanner to plan a collision-free path.

Messages: These represent the specific data packets (e.g. strings, integers, floats or complex sensor structures like `LaserScan` or `BatteryState`) that are sent out by ~~publishers~~ publishers.

Topics: These are the named channels or "buses" over which messages are transmitted. By convention, topic names start with a forward slash (e.g. /scan), allowing nodes to communicate by publishing to or subscribing from these established channels.

Steps for writing the code for a ROS2 Python node:

- 1) Navigate to your source directory of your workspace
- 2) Open VS code to write the script (code.)
- 3) Start with shebang line (`#!/usr/bin/env python3`)
- 4) Import necessary libraries (rcipy, node class & specific message type - `std_msgs.msg import String`)
- 5) Define a class for your node that inherits from `Node`
- 6) Create the publisher
- 7) Define a main function
  - Create an `__init__.py` file
  - update the xml file
  - configure `CMakeLists.txt`

Build & Run

Build

`ros2 run <pkg name> <node>`

`ROS2 topic list`

→ lists active topics

`ros2 topic echo /<topic name>`

→ shows contents of topic

`ros2 topic hz /py-example-topic`

→ shows no. of topics published per second

`ros2 topic info /py-example-topic`

→ more info on the topic like type, publishers & subscribers

`ros2 node list`

→ lists active nodes

`ros2 node info <node name>`

→ info on node

`rqt_graph`

→ visualizes how the nodes are communicating with each other